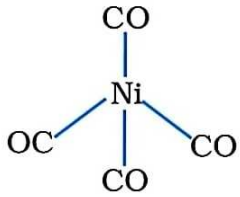
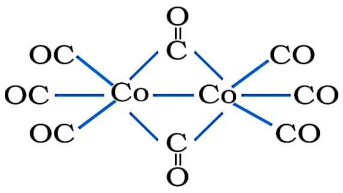
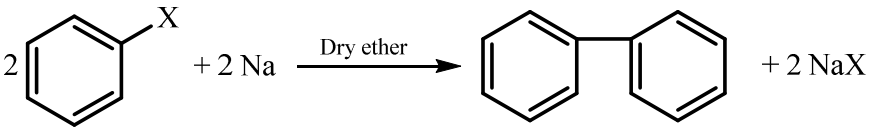
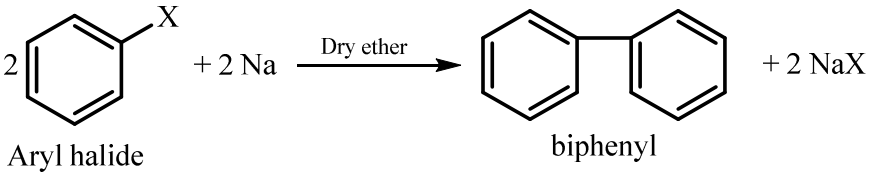
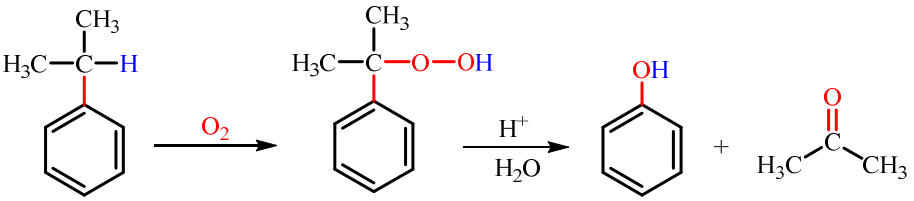
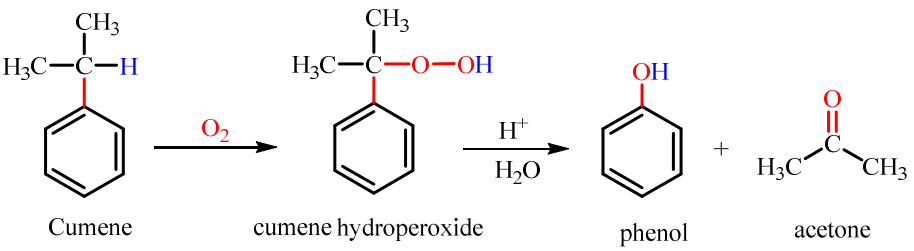
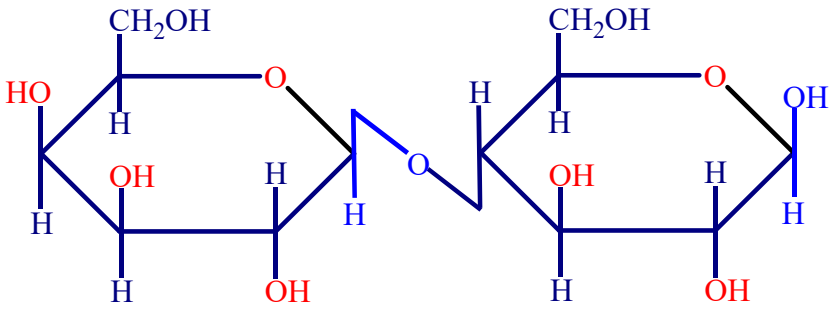
KARNATAKA SCHOOL EXAMINATION & ASSESSMENT BOARD
II PUC EXAM – 2 APRIL/MAY– 2025

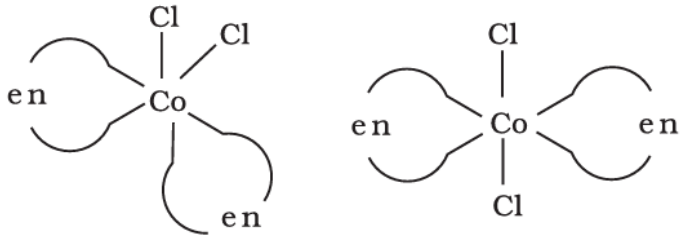
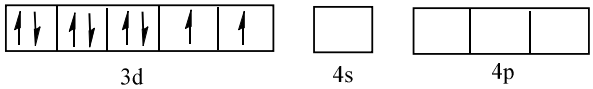
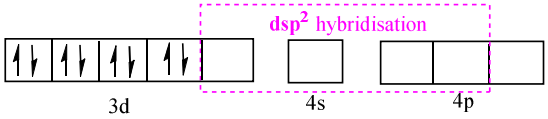
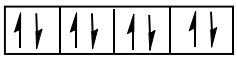
SUBJECT: 34-CHEMISTRY MODEL ANSWERS MAX. MARKS: 70

	PART-A	
I.	**Select the correct option from the given choices:** **15 × 1=15**	
1)	When compared to ΔT_f -value of a **0.01M** glucose solution, the ΔT_f - value of **0.01M $MgCl_2$** solution a) increases about three times b) increases about six times c) increases about ten times d) remains same	
Ans:	**a) increases about three times (or) a) (or) increases about three times**	**1**
2)	The product/s obtained during the electrolysis of aqueous sodium chloride solution is/are a) H_2 only b) Na metal only c) Na and Cl_2 d) H_2, Cl_2 and NaOH	
Ans:	**d) H_2, Cl_2 and NaOH (or) d) (or) H_2, Cl_2 and NaOH**	**1**
3)	All natural and artificial radioactive decay of unstable nucleii take place by a) zero order kinetics b) first order kinetics c) second order kinetics d) third order kinetics	
Ans:	**b) first order kinetics (or) b) (or) first order kinetics**	**1**
4)	The energy required to form activated complex is known as a) Kinetic energy b) Potential energy c) Activation energy d) Collision frequency	
Ans:	**c) Activation energy (or) c) (or) Activation energy**	**1**
5)	Which of the following statements is NOT correct? a) $KMnO_4$ oxidises iodide ion to iodine in acid medium b) Permanganate is diamagnetic c) Copper liberates hydrogen from acids d) Ti^{2+} liberates hydrogen from dilute acids	
Ans:	**c) Copper liberates hydrogen from acids (or) c) (or) Copper liberates hydrogen from acids**	**1**
6)	The most stable complex in the following complexes is a) $[Fe(CO)_5]$ b) $[(Fe(CN)_6]^{3-}$ c) $[Fe(H_2O)_6]^{3+}$ d) $[Fe(C_2O_4)_3]^{3-}$	
Ans:	**(d) $[Fe(C_2O_4)_3]^{3-}$ (or) (d) (or) $[Fe(C_2O_4)_3]^{3-}$**	**1**
7)	Match the List - I(complex) with List - Il (central metal) and select the correct option List I List II A. Blood pigment i) Rh B. Chlorophyll ii) Co C. Vitamin B_{12} iii) Mg D. Wilkinson Catalyst iv) Fe a) A - i, B - ii, C - iii, D - iv b) A - iv, B - iii, C - ii, D – i c) A - iv, B - ii, C - iii, D - i d) A - i, B - iii, C- ii, D - iv	
Ans:	**b) A - iv, B - iii, C - ii, D – i (or) b) (or) A - iv, B - iii, C - ii, D – i**	**1**
8)	The main product obtained when alkyl halides react with AgCN is a) cyanide b) isocyanide c) alkyl nitrite d) nitroalkane	
Ans:	**b) isocyanide (or) b) (or) isocyanide**	**1**

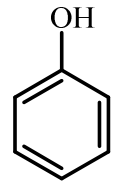
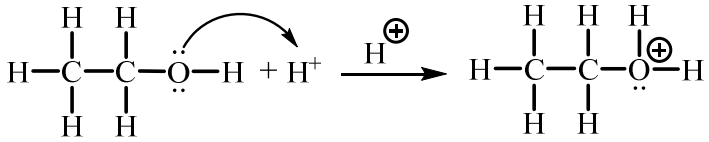
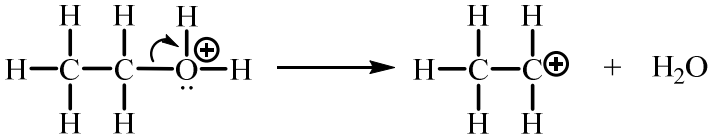
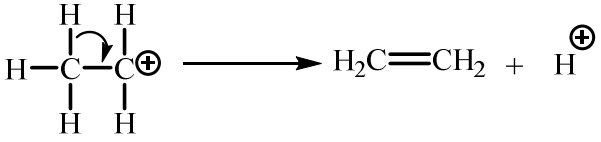
9)	The compound used for the treatment of malaria is a) chloroquine b) chlorophenicol c) halothane d) chloroform	
Ans:	**a) chloroquine (or) a) (or) chloroquine**	**1**
10)	Which of the following is benzylic alcohol? a) $C_6H_5—CH_2—CH_2—OH$ b) $C_6H_5—CH_2—CH(CH_3)—OH$ c) $C_6H_5—CH_2—OH$ d) $H_3C—CH_2—OH$	
Ans:	**c) C_6H_5-CH_2OH (or) c) (or) C_6H_5-CH_2OH**	**1**
11)	Among the following, the compound with highest pK_a value is a) o-nitro phenol b) phenol c) p-nitro phenol d) ethanol	
Ans:	**d) ethanol (or) d) (or) ethanol**	**1**
12)	In Clemmensen reduction, carbonyl compound is treated with a) Zinc amalgam + conc. HNO_3 b) Zinc amalgam + conc. HCl c) Sodium amalgam + conc. HNO_3 d) Sodium amalgam + conc. HCl	
Ans:	**b) Zinc amalgam + conc. HCl (or) b) (or) Zinc amalgam + conc. HCl**	**1**
13)	The reagent that can be used to convert benzene diazonium chloride to benzene is a) $H_3PO_2 + H_2O$ b) $SnCl_2/HCl$ c) $LiAlH_4$ d) H_3PO_4	
Ans:	**a) $H_3PO_2 + H_2O$ (or) a) (or) $H_3PO_2 + H_2O$**	**1**
14)	The final product obtained in ammonolysis of alkyl halides is a) quaternary ammonium salt b) primary amine c) secondary amine d) tertiary amine	
Ans:	**a) quaternary ammonium salt (or) a) (or) quaternary ammonium salt**	**1**
15)	Statement-I: All naturally occurring α-amino acids except glycine are optically active. Statement-II: Most naturally occurring aminoacids have 'D' configuration. Identify the correct option a) Statement-I and Statement-II are correct b) Statement-I and Statement-II are incorrect c) Statement-I is correct and Statement-II is incorrect d) Statement-I is incorrect and Statement-II is correct	
Ans:	**c) Statement-I is correct and Statement-II is incorrect (or) c) (or) Statement-I is correct and Statement-II is incorrect**	**1**
II.	**Fill in the blanks by choosing the appropriate word from those given in the brackets: [SCN^-, Cellulose, Starch, Gibb's energy, benzene, nylon-6,6]**	**5×1=5**
16)	A catalyst does not alter _____.	
Ans:	**Gibb's energy**	**1**
17)	_____ is an example of ambidentate ligand.	
Ans:	**SCN^-**	**1**
18)	Phenol is converted to _____ on heating with zinc dust.	
Ans:	**benzene**	**1**
19)	Adipic acid is used in the manufacture of _____.	
Ans:	**nylon-6,6**	**1**
20)	The most abundant organic substance in plant kingdom is _____.	
Ans:	**Cellulose**	**1**

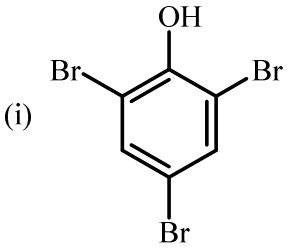
	PART - B	
III.	**Answer any three of the following. Each question carries 2 marks.**	**3 × 2 = 6**
21)	**What are azeotropes? Give an example.**	
Ans:	Azeotropes are binary mixtures having the same composition in liquid and vapour phase and boils at a constant temperature. **(or)** A solution which distills over without change in composition at constant temperature. **Example:** 95% ethanol **(or)** 68% HNO_3 + 32% H_2O **(or)** phenol and aniline **(or)** chloroform and acetone **(or)** **(Any other suitable example)**	**1** **1**
22)	**Draw the structures of $[Ni(CO)_4]$ and $[Co_2(CO)_8]$**	
Ans:	 $Ni(CO)_4$  $[Co_2(CO)_8]$	**1+1**
23)	**Explain Fittig reaction with an example**	
Ans:	Aryl halides when treated with sodium in dry ether **give biphenyl (diphenyl).**  **(or)**  Aryl halide biphenyl **(Any suitable chemical equation; statement 1 mark and equation 1 mark (or) self-explanatory equation 2 mark)**	**1** **1** **(or)** **2**
24)	**How do you convert isopropyl benzene into phenol? Write equations.**	
Ans:	Isopropylbenzene (cumene) is oxidized in the presence of air to form cumene hydroperoxide, followed by hydrolysis gives phenol and acetone.  **(or)**  Cumene cumene hydroperoxide phenol acetone	**1** **1** **(or)** **2**

25)	**Write the Haworth structure of lactose.**	
Ans:	 Lactose	**2**
	PART - C	
IV.	**Answer any three of the following. Each question carries 3 marks.**	**3 × 3 = 9**
26)	**What are interstitial compounds? Write any two characteristics of them.**	
Ans:	Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.	**1**
	1) They have high melting points, higher than those of pure metals. 2) They are very hard. 3) They retain metallic conductivity. 4) They are chemically inert. **(Any two one mark for each)**	**2**
27) a)	**Calculate the magnetic moment of Fe^{2+} by using spin only formula (Atomic no. of Fe = 26)**	
b)	**Write the equation for the reaction of lanthanoids with oxygen.**	
Ans: a)	$\mu = \sqrt{n(n+2)}$ BM $= \sqrt{4(4+2)}$ $= \sqrt{24}$ BM (or) $= 4.89$ BM	**1** **1**
b)	$4Ln + 3O_2 \longrightarrow 2Ln_2O_3$ **(or)** $Ln \xrightarrow{\text{burns in } O_2} Ln_2O_3$	**1**
28)	**Give reason for the following:**	
a)	**Zn^{+2} is colourless.**	
b)	**Cerium shows +4 oxidation state.**	
c)	**Zr and Hf exhibit similar radii.**	
Ans: a)	Due to the absence of unpaired electron **(or)** Due to pairing of all electrons **(or)** Due to completely filled 3d sub shell **(or)** $3d^{10}$ configuration.	**1**
b)	Due to attaining the noble gas configuration (Xe) **(or)** Due to attaining $4f^0$ configuration	**1**
c)	Due to lanthanoid contraction	**1**

29)	**Write the IUPAC name of $[CoCl_2(en)_2]$. Draw the geometrical isomers of it.**	
Ans:	Dichloridobis(ethane-1,2-diamine)cobalt(II)  	**1** **1+1**
30)	**Explain the hybridisation, geometry and magnetic property of $[Ni(CN)_4)]^{2-}$, on the basis of valence bond theory (VBT). (Atomic number of Ni= 28)**	
Ans:	In $[Ni(CN)_4]^{2-}$; the Nickel ion is in +2 oxidation state. Electronic configuration of Ni^{2+}: [Ar] $3d^8$ $4s^0$  When four **CN^-** ligands approaching central metal ion, rearrangement of electrons takes place.  One d-orbital, one s-orbital and two p-orbitals undergo hybridization to form four dsp^2 hybrid orbitals of Nickel.  Four dsp^2 hybrid orbitals These four hybridised orbitals of Ni^{2+} form four coordinate bonds with cyanide ligands.  Four dsp^2 hybrid orbitals four pair of electrons of cyanide ligand The complex has square planar geometry. diamagnetic because of the absence of unpaired electron. **(or)** **Hybridisation:** dsp^2. **Magnetic property:** Diamagnetic. **Geometry:** Square planar.	**Geometry:** one-mark, **Magnetic property:** one mark; **Explanation:** one mark
V	**Answer any two of the folowing. Each question carries 3 marks:**	**2 × 3 = 6**
31) a) **b)**	**State Henry's law. Write its mathematical form.** **What is the effect of pressure on solubility of solids in liquids?**	
Ans: a)	Henry's law states that at a constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas present above the surface of liquid or solution. **(or)** "The partial pressure of the gas in vapour phase is proportional to mole fraction of the gas in the solution" **(or)** "Mole fraction of gas in the solution is proportional to the partial pressure of the gas over the solution" **Mathematical form:** $p = K_H \chi$	**1** **1**
b)	No Change **(or)** No effect **(or)** un effected	**1**

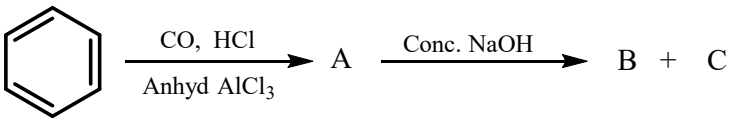
32)	**Write the cathodic, anodic and overall reactions occur in fuel cell using H_2 and O_2.**	
Ans:	**Cathode:** $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$	**1**
	Anode: $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$	**1**
	Overall reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	**1**
33) a)	**Define molar conductivity. Write its SI unit.**	
b)	**How much electricity (in terms of Faraday) is required for the oxidation of FeO to Fe_2O_3?**	
Ans: a)	Conductance of a solution containing one mole of an electrolyte. **(or)** Conductance due to all the ions produced from one mole of an electrolyte dissolved **(or)** Conductance of the solution containing one mole of an electrolyte kept between two electrodes unit length apart and the area of the electrodes is so large that the whole of the solution is contained between them.	**1**
	$Sm^2\ mol^{-1}$	**1**
b)	1F **(or)** 1 Faraday **(or)** One Faraday	**1**
34)	**Derive an integrated rate equation for the rate constant of a zero-order reaction.**	
Ans:	Consider a zero-order reaction, $R \rightarrow P$ Let $[R]_0$ be initial concentration of the reactant and [R] be concentration of the reactant at any time "t". $$\text{Rate} = \frac{-d[R]}{dt} = k[R]^0$$	**1**
	Where k is rate constant of a zero-order reaction $$\text{Rate} = \frac{-d[R]}{dt} = k \times 1$$ $$d[R] = -k\ dt$$	**1**
	Integrating both sides: $[R] = -kt + I$ … (i) (I = Integration constant) When $t = 0$, $[R] = [R]_0$; Substituting in equation (i), $$[R]_0 = -k \times 0 + I$$ $$[R]_0 = I$$ Substituting value of 'I' in equation (i), $$[R] = -kt + [R]_0$$ $$k = \frac{[R]_0 - [R]}{t}$$	**1**

	PART - D	
VI.	**Answer any four of the following. Each question carries 5 marks.**	**4 × 5 = 20**
35) a)	**Aryl halides are less reactive than alkyl halides towards nucleophilic substitution reactions. Give any three reasons for it.**	
b)	**Write the general equation for Finkelstein reaction. What is the role of dry acetone in this reaction?**	
Ans: a)	1. Resonance effect 2. Difference in hybridization of carbon atom in C-X bond 3. Instability of phenyl cation 4. Repulsion between electron rich nucleophile and electron rich arenes **(Any three, one mark for each)**	**3**
b)	$R{-}X + NaI \xrightarrow{\text{Dry Acetone}} R{-}I + NaX$ $X = Cl, Br$	**1**
	Dry acetone facilitates the forward reaction **(or)** Serves as both a solvent and a drying agent **(or)** Facilitating the S_N2 reaction by dissolving reactants **(or)** Driving the reaction forward by promoting precipitation of the by-product sodium halide (NaX).	**1**
36) a)	**Write the equations for the steps in the mechanism of acid catalyzed dehydration of ethanol to ethene.**	
b)	**Complete the following equations:**	
	(i)  $+ 3\,Br_2 \longrightarrow$ ——— $+ 3\,HBr$	
	(ii) $C_2H_5Br + C_2H_5ONa \longrightarrow$ ——— $+ NaBr$	
Ans: a)	**STEP-1:** 	**1**
	STEP-2: 	**1**
	STEP-3: 	**1**
	(or)	**(or)**
	$CH_3CH_2{-}OH + H^+ \longrightarrow CH_3CH_2{-}\overset{+}{O}H_2$	**1**
	$CH_3CH_2{-}\overset{+}{O}H_2 \longrightarrow CH_3\overset{+}{C}H_2 + H_2O$	**1**
	$CH_3\overset{+}{C}H_2 \longrightarrow H_2C{=}CH_2 + H^+$	**1**

b) (i) 

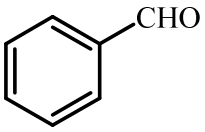
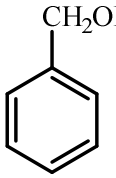
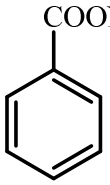
(ii) $C_2H_5OC_2H_5$ — **1+1**

37) a) Identify the compounds A, B and C in the following equation:



b) What is the composition of Tollens' reagent? Write the reaction of it with aldehydes in alkaline medium.

Ans: a)

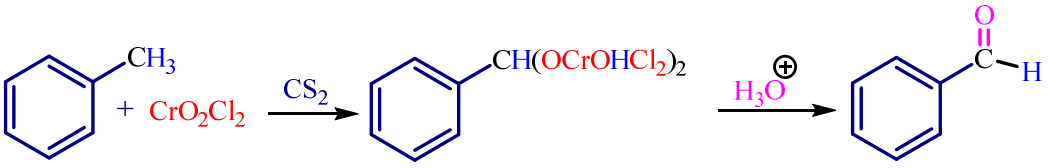
A	 **(or)** C_6H_5CHO **(or)** Benzaldehyde	**1**
B	 **(or)** $C_6H_5CH_2OH$ **(or)** Benzyalalcohol	**1**
C	 **(or)** C_6H_5COONa **(or)** Sodium benzoate	**1**

b) Ammonical silver nitrate **(or)** Ammonical $AgNO_3$ — **1**

$RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag + 2H_2O + 4NH_3$ — **1**

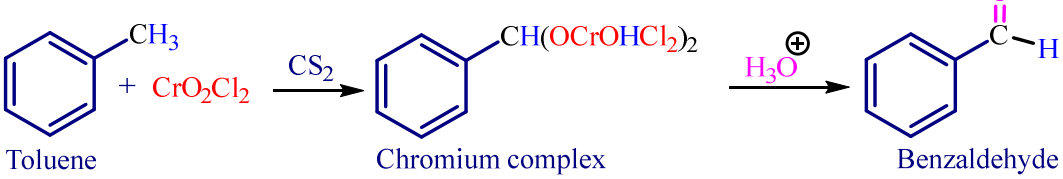
38) a) Explain Etard reaction with equation.
b) How will you prepare acetamide from acetic acid? Write the equations.
c) Between HCHO and CH_3CH_2CHO, which is the compound undergoing aldol reaction?

Ans: a) When toluene is treated with chromyl chloride forms a chromium complex, which on hydrolysis gives benzaldehyde. — **1**



1

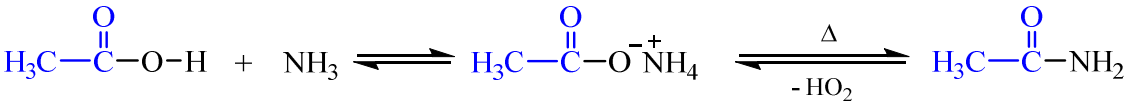
(or)



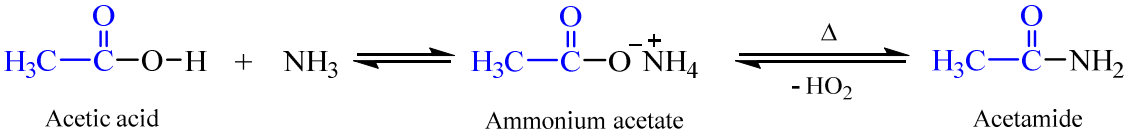
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(Marks should be awarded for reaction without intermediate) — **1**

b) Acetic acid reacts with ammonia to give ammonium acetate which on further heating at high temperature give acetamide. **1**



(or) **(or)**



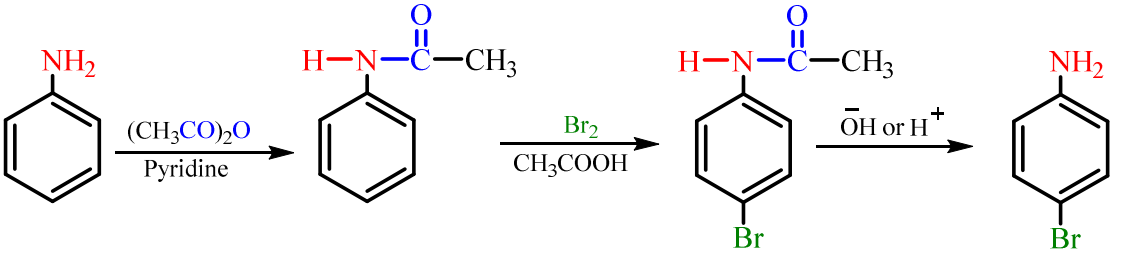
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(Any alternative correct preparation methods and Marks should be awarded for reaction without intermediate)

c) CH_3CH_2CHO **1**

39 a) Write the equations for the conversion of aniline into 4-bromo aniline.
b) How do you distinguish between primary amine and secondary amine using Hinsberg's reagent?

Ans: a)

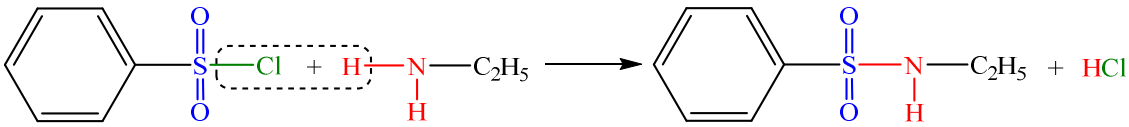


3

(One mark for each step)

b) Primary amines reacts with benzenesulphonyl chloride gives precipitate (*N*-ethylbenzenesulphonyl amide) which is soluble in alkali. **1**

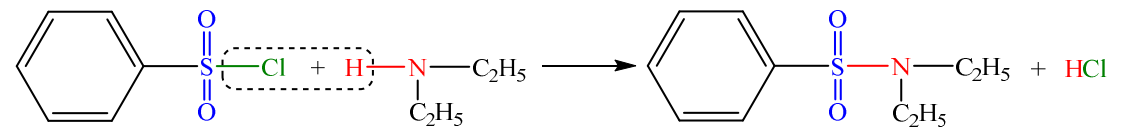
(or) **(or)**



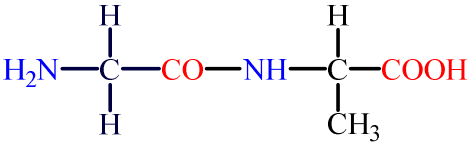
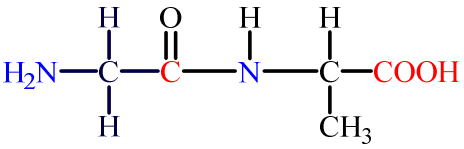
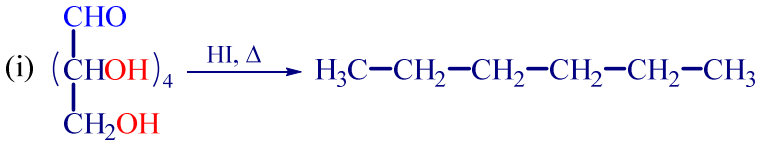
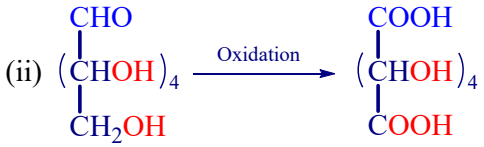
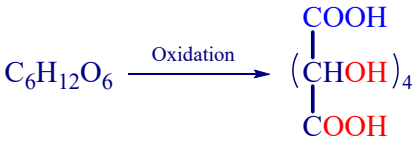
1

Secondary amines reacts with benzenesulphonyl chloride gives precipitate (*N*,*N*-diethylbenzene sulponamide), which is insoluble in alkali **1**

(or) **(or)**



1

40) a) **b)** **c)**	**What is peptide linkage? Write the structure of glycylalanine.** **Write the chemical reactions to show that glucose contains** **(i) Six carbon atoms in a straight chain.** **(ii) Primary alcoholic (-OH) group.** **Name the disease caused by the deficiency of ascorbic acid.**	
Ans: a)	Peptide linkage is a bond formed between -COOH group and $-NH_2$ group **(or)** –CO-NH- bond between two α-aminoacids.  **(or)** 	**1** **1**
b)	(i)  **(or)** (i) $C_6H_{12}O_6 \xrightarrow{HI,\ \Delta} C_6H_{14}$ (ii)  **(or)**  **(Oxidation takes place with Nitric acid,)**	**1** **(or)** **1** **1**
c)	Scurvy	**1**
	PART – E (PROBLEMS)	
VII.	**Answer any three of the following. Each question carries 3 marks.**	**3 × 3 = 9**
41)	**Calculate the mole fraction of benzene and carbon tetrachloride in a solution made by dissolving 20g of benzene in 80g of carbon tetrachloride.** **(Molar mass of benzene =78g mol^{-1}, Molar mass of carbon tetrachloride =154g mol^{-1})**	
Ans:	$\chi_{C_6H_6} = \frac{n_{C_6H_6}}{n_{C_6H_6} + n_{CCl_4}}$ $\chi_{CCl_4} = \frac{n_{CCl_4}}{n_{CCl_4} + n_{C_6H_6}}$ $= \frac{0.2564}{0.2564+0.5194}$ (or) $= \frac{0.5194}{0.5194+0.2564}$ $\chi_{C_6H_6} = 0.3304$ $\chi_{CCl_4} = 0.6703$	**1** **1** **1**

42)	**Determine the amount (in grams) of $CaCl_2$ (i =2.47) dissolved in 2.5L of water such that its osmotic pressure is 0.75 atm at 27°C.** **(R = 0.0821 L.atm.K^{-1}.mol^{-1}, molar mass of $CaCl_2$ = 111 g.mol^{-1})**	
Ans:	$\pi = iCRT \quad (or) \quad \pi = i\frac{n}{V}RT \quad (or) \quad \pi = i\frac{W_B}{M_B}\times\frac{RT}{V} \quad (or) \quad W_B = \frac{\pi M_B V}{iRT}$ $W_B = \frac{0.75\times111\times2.5}{2.47\times0.0821\times300}$ $W_B = 3.42\,g$ **(or)** $\pi = iCRT \quad (or) \quad \pi = i\frac{n}{V}RT \quad (or) \quad n = \frac{\pi V}{iRT}$ $n = \frac{0.75\times2.5}{2.47\times0.0821\times300}$ Amount of $CaCl_2$ = 0.0308×111 = 3.4188 g (or) = 3.42 g **(Strictly expecting correct unit with final answer)**	**1** **1** **1** **(or)** **1** **1** **1**
43)	**At 25°C, the following reaction takes place in the cell which has E^0_{cell} = 0.235V. Calculate the standard Gibb's energy (F = 96500 $Cmol^{-1}$).** **$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + I_{2(s)}$**	
Ans:	$\Delta_r G^o = -nFE^o_{Cell}$ $\therefore \Delta_r G^o = -2\times96500\times0.235$ $\Delta_r G^o = -45355\ J\ mol^{-1}$ **(or)** $\Delta_r G^o = -45.355\ kJ\ mol^{-1}$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**
44)	**A solution of $Ni(NO_3)_2$ is electrolyzed between platinum electrodes using a current of 5A for 20 min. What mass of Ni is deposited at the cathode?** **(Atomic mass of Ni =58.7, F=96500 C mol^{-1})**	
Ans:	Q = It = 5×20×60= 6000 C $Ni(NO_3)_2 \longrightarrow Ni^{2+} + 2NO_3^-$ $Ni^{2+} + 2e^- \longrightarrow Ni$ 2F 1mole(58.7 g) $\therefore 2\times96500\,C$ deposits 58.7g of Ni $6000\,C \text{ deposits} = \frac{58.7\times6000}{2\times96500} = 1.8248\,g$ **(or)** $W = \frac{\text{Molecular weight}}{n\times F}\times it$ $= \frac{58.7}{2\times96500}\times5\times20\times60$ $= 1.8248\,g$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1** **(or)** **1** **1** **1**

45)	**A first order reaction takes 40 min for 25% completion. Calculate rate constant of the reaction.**	
Ans:	$k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$ $k = \frac{2.303}{40}\log\frac{100}{75}$ (or) $k = \frac{2.303}{40\times60}\log\frac{100}{75}$ $k = 0.007191\,min^{-1}$ (or) $= 7.19\times10^{-3}\,min^{-1}$ (or) $= 0.00011985\,s^{-1}$ (or) $= 1.1985\times10^{-4}\,s^{-1}$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**
46)	**The rate constant for the first order decomposition of H_2O_2 is given by the following equation: log k=14.34 - 1.25 x 10^4 K/T** **Calculate E_a for the reaction (R=8.314 J.K- mol^{-1})**	
Ans:	Given; $\log k = 14.34 - 1.25\times10^4\frac{K}{T}$ $\log k = \log A - \frac{E_a}{2.303RT}$ $\frac{E_a}{2.303RT} = \log A - \log k$ $\frac{E_a}{2.303RT} = 14.34 - 14.34 + \frac{1.25\times10^4}{T}$ $\frac{E_a}{2.303R} = 1.25\times10^4$ $E_a = 1.25\times10^4\times2.303\times8.314$ $E_a = 239339.275\,J\,mol^{-1}$ (or) $= 239.339\,kJ\,mol^{-1}$ (or) $= 239.34\,kJ\,mol^{-1}$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**

*********END*********